Round-trip closure on directed k-nearest-neighbour graphs for multiscale clustering

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Abstract

We introduce a graph-based clustering formulation based on recursive round-trip closure on directed $\boldsymbol{k}$-nearest-neighbour graphs. The method is motivated by an earlier relational model in which an asymmetric relation $\boldsymbol{T} \subseteq \boldsymbol{A} \times \boldsymbol{B}$ induces a reflective operator $\boldsymbol{R_T} = \boldsymbol{T^{-1}} \circ \boldsymbol{T}$, and by a subsequent KNN-based model in which each nearest neighbour can reveal its reverse-neighbour set. In the present formulation, a data point first reaches one or more of its $\boldsymbol{K}$ nearest neighbours, and each selected neighbour returns the set of data points that also selected it as one of their $\boldsymbol{K}$ nearest neighbours. This forward-backward operation induces local popularity, leadership, and recursive expansion until saturation. Repeating the process across increasing values of $\boldsymbol{K}$ produces a multiscale sequence of cluster layers, allowing cluster persistence, absorption, and merging to be analysed through overlap transitions. The method is positioned as a nearest-neighbour graph clustering variant combining directed KNN topology, reverse-neighbour popularity, round-trip closure, saturation depth, and $\boldsymbol{K}$-layer analysis. Preliminary synthetic results indicate that the generated layers are valid partitions, with no internal overlap among cluster memberships for each analysed value of $\boldsymbol{K}$.

Keywords: k-nearest-neighbour graph, graph-based clustering, reverse nearest neighbours, shared nearest neighbours, round-trip closure, multiscale clustering, saturation, cluster hierarchy

1 Introduction

Nearest-neighbour relations are among the most fundamental structures used to represent finite datasets embedded in metric spaces. They occur naturally in pattern classification, nonparametric density estimation, graph construction, manifold learning, outlier detection, and clustering [1–5]. In clustering, k-nearest-neighbour (KNN) relations have been explored through shared nearest neighbours, reciprocal nearest neighbours, mutual KNN graphs, reverse nearest-neighbour statistics, and density-based graph connectivity [6–11].

This paper develops a clustering formulation based on a recursive *round-trip* operation in directed KNN graphs. Given a data point x_i, the forward step reaches one of its K nearest neighbours, say $x_j \in \mathcal{N}_K^+(x_i)$. The backward step then retrieves all data points that also selected x_j as one of their K nearest neighbours:

$$\mathcal{N}_K^-(x_j) = \{x_\ell \in X : x_j \in \mathcal{N}_K^+(x_\ell)\}. \tag{1}$$

Thus, a neighbour is not used merely as a point in Euclidean space, but as an index to a reverse-neighbourhood set. This operation can be recursively applied until no additional points are reached, producing a saturated closure.

The method has two historical sources. The first is a relational model of redundancy-based information transport, in which an asymmetric relation $T \subseteq A \times B$ induces a reflective composition $T^{-1} \circ T$. The second is a KNN-based clustering model, where a data point elects a locally popular neighbour by inspecting reverse-neighbour cardinalities. The present contribution is not the isolated use of KNN graphs, reverse nearest neighbours, or shared-neighbour similarity. Rather, it is the organisation of these elements into a recursive closure process whose evolution across K values defines a multiscale cluster-layer structure.

1.1 Novelty and positioning

The novelty of the proposed method is not the isolated use of KNN graphs, reverse-neighbour counts, or shared-neighbour evidence. These components are well established in the nearest-neighbour clustering literature. The contribution lies in their organisation into a recursive round-trip closure operator on directed KNN graphs, whose saturated outputs induce scale-indexed partition layers. The resulting K-layer sequence provides a multiscale description of cluster formation, absorption, and merging without relying on a classical linkage criterion or on the minimisation of an explicit objective function.

In this sense, the method is best understood as an algorithmic and topological construction. It does not reduce to a weighted voting rule, nor to the particular choice of a distance-based weight such as an inverse-square law. The central object is the dynamics induced on the directed KNN graph: forward-neighbour selection, reverse-neighbour retrieval, popularity, visibility, recursive expansion, and saturation.

1.2 Main contributions

The main contributions of this paper are as follows:

1. a formal round-trip operator on directed KNN graphs based on forward nearest-neighbour selection and backward reverse-neighbour retrieval;
2. a popularity-based local leadership mechanism using reverse-neighbour cardinality;
3. a recursive closure process with an explicit saturation criterion;
4. a K-layer representation for multiscale cluster evolution;
5. an overlap-based diagnostic for verifying whether layers are proper partitions and for studying cluster absorption and merging across K.

2 Related work

2.1 Shared-neighbour and Jarvis–Patrick clustering

Jarvis and Patrick introduced a clustering criterion based on shared near neighbours [6]. Later shared-nearest-neighbour (SNN) clustering methods used common neighbour counts to improve robustness to clusters of different shapes, sizes, and densities [7]. The present method is related to this family because reverse-neighbourhood intersections imply shared-neighbour evidence. However, shared-neighbour information is not used only as a static similarity score; it is embedded in a recursive graph-closure process.

2.2 Mutual, reciprocal, and reverse nearest-neighbour methods

Mutual KNN and reciprocal nearest-neighbour graphs have been used for clustering and outlier detection [8, 9, 12]. Reverse nearest-neighbour methods use the number of incoming KNN links as a density or representativeness signal [10]. In the present formulation, the reverse degree acts as a local popularity measure and can determine leadership within a point's KNN list.

2.3 Density-based and spectral graph clustering

DBSCAN, OPTICS, and HDBSCAN define clusters through density connectivity and hierarchical density estimates [13–16]. Spectral clustering constructs a similarity graph and uses eigenvectors of the associated graph Laplacian to reveal cluster structure [5, 17]. The method proposed here is graph-based, but it is combinatorial and closure-driven: it does not require eigenvalue decomposition and it exposes a sequence of partitions as the neighbourhood size K varies.

3 Relational origin of the round-trip operator

The primitive idea can be stated for an arbitrary directed relation. Let

$$A = \{a_1, \ldots, a_N\}, \qquad B = \{b_1, \ldots, b_M\} \tag{2}$$

be finite sets and let

$$T \subseteq A \times B \tag{3}$$

be an asymmetric relation. The reflective composition induced by T is

$$R_T = T^{-1} \circ T \subseteq A \times A. \tag{4}$$

For $a \in A$, the reflective neighbourhood is

$$R_T(a) = T^{-1}(T(a)). \tag{5}$$

Equation (5) means that one first moves from a to the objects related to it in B and then returns to all objects in A that share those objects in B.

For a subset $S \subseteq A$, define

$$R_T(S) = \bigcup_{a \in S} R_T(a). \tag{6}$$

The recursive reflection is

$$R_T^0(S) = S, \tag{7}$$

$$R_T^{t+1}(S) = R_T(R_T^t(S)), \qquad t \geq 0. \tag{8}$$

The process saturates when

$$R_T^{s+1}(S) = R_T^s(S) \tag{9}$$

for the first time. The integer s is the saturation depth of S under R_T.

This relational view is not the algorithmic core of the present paper, but it provides the mathematical origin of the forward-backward idea. The algorithmic core is obtained by instantiating the relation on a directed KNN graph.

4 Directed KNN graphs and reverse neighbourhoods

Let

$$X = \{x_1, \ldots, x_N\} \subset \mathbb{R}^d \tag{10}$$

be a finite dataset with a metric $\delta : X \times X \to \mathbb{R}_{\geq 0}$. For each point x_i, let

$$\mathcal{N}_K^+(x_i) = \mathrm{KNN}_K(x_i) \tag{11}$$

denote the set of its K nearest proper neighbours. The directed KNN graph is

$$\mathcal{G}_K = (X, E_K), \tag{12}$$

where

$$(x_i, x_j) \in E_K \quad \Longleftrightarrow \quad x_j \in \mathcal{N}_K^+(x_i). \tag{13}$$

The reverse-neighbour set of x_j is

$$\mathcal{N}_K^-(x_j) = \mathrm{RNN}_K(x_j) = \{x_i \in X : x_j \in \mathcal{N}_K^+(x_i)\}. \tag{14}$$

The reverse degree, or popularity, is

$$\mathrm{rdeg}_K(x_j) = |\mathcal{N}_K^-(x_j)|. \tag{15}$$

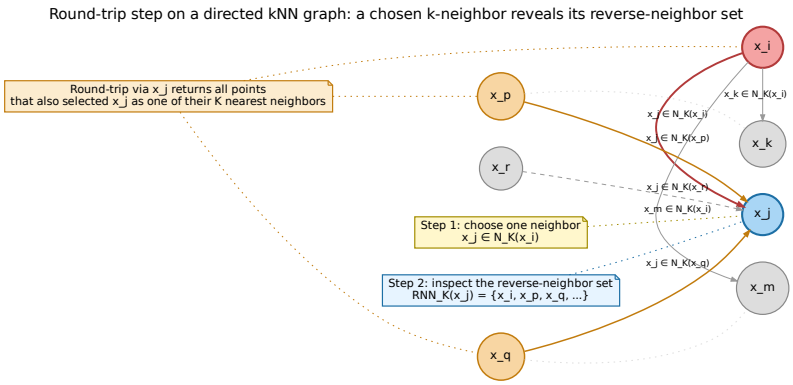


Fig. 1 Operational round-trip step on a directed KNN graph. A point x_i selects x_j as one of its K nearest neighbours. The selected neighbour x_j then reveals its reverse-neighbour set, i.e., all points that also selected x_j as one of their K nearest neighbours.

Figure 1 illustrates the elementary operation. The forward step is a standard KNN edge. The backward step is not a Euclidean displacement; it is a query to the reverse-neighbour set of the selected point. This is the local mechanism that links the 2018 KNN model to the older reflective composition $T^{-1} \circ T$.

Definition 1 (Round-trip neighbourhood)**.** *Given $x_i \in X$, its one-step round-trip neighbourhood is*

$$\mathcal{R}_K(x_i) = \bigcup_{x_j \in \mathcal{N}_K^+(x_i)} \mathcal{N}_K^-(x_j). \tag{16}$$

Equation (16) can be restricted by selecting only one neighbour, for example the most popular neighbour in $\mathcal{N}_K^+(x_i)$, or expanded by allowing all forward neighbours to contribute.

5 Popularity-based leadership

The 2018 formulation emphasised that a neighbour should be evaluated not only by its distance to the query point, but also by how many other points selected it as a nearest neighbour. This leads to a local leadership mechanism.

Definition 2 (Local leader)**.** *The popularity leader of x_i at scale K is*

$$\ell_K(x_i) = \operatorname*{arg\,max}_{x_j \in \mathcal{N}_K^+(x_i)} \operatorname{rdeg}_K(x_j). \tag{17}$$

The leader $\ell_K(x_i)$ is one of the K nearest neighbours of x_i, but it is chosen by reverse-neighbour popularity. Ties can be resolved by distance, by index order, or by retaining multiple leaders.

The leadership graph is a compressed directed structure obtained from the original KNN graph. It is not required that leadership alone define the final clustering, but

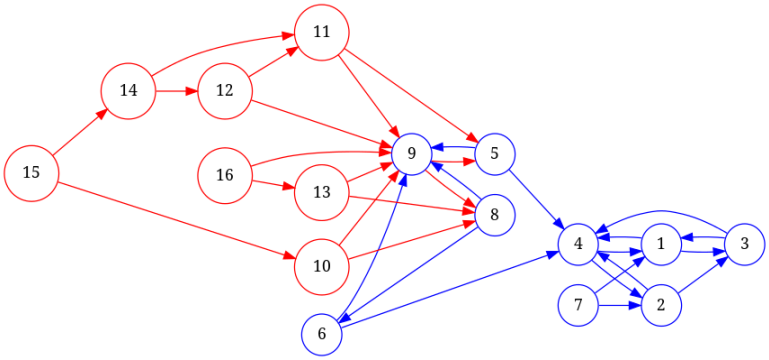


Fig. 2 Example of a directed KNN graph from the 2018 prototype. Edges represent nearest-neighbour choices. The same graph supports reverse-neighbour counting and local leadership.

it provides a useful mechanism for directing the round-trip expansion toward locally representative points.

6 Recursive round-trip closure and saturation

The simplest closure version uses all forward neighbours:

$$C_K^0(x_i) = \{x_i\}, \tag{18}$$

$$C_K^{t+1}(x_i) = C_K^t(x_i) \cup \bigcup_{x_q \in C_K^t(x_i)} \mathcal{R}_K(x_q). \tag{19}$$

The process saturates when

$$C_K^{t+1}(x_i) = C_K^t(x_i). \tag{20}$$

The saturated closure is denoted

$$C_K^\infty(x_i) = C_K^s(x_i), \tag{21}$$

where s is the first saturation time.

A leadership-driven variant is

$$C_K^{t+1}(x_i) = C_K^t(x_i) \cup \bigcup_{x_q \in C_K^t(x_i)} \mathcal{N}_K^-(\ell_K(x_q)). \tag{22}$$

This version follows the reverse-neighbour set of the most popular local leader of each currently reached point.

The clustering step can then merge points whose saturated closures coincide or overlap above a prescribed threshold. In the current implementation, clusters are obtained from the induced layer output and then analysed as partitions.

Algorithm 1 Leadership-driven round-trip closure for one point

Require: directed KNN graph $\mathcal{G}_K$, point x_i

$C \leftarrow \{x_i\}$
repeat
  $C_{old} \leftarrow C$
  for all $x_q \in C_{old}$ **do**
    $y \leftarrow \ell_K(x_q)$
    $C \leftarrow C \cup \mathcal{N}_K^-(y)$
  end for
until $C = C_{old}$
return C

7 Cluster layers and multiscale evolution

For a fixed K, the method yields a partition or cluster layer

$$\mathcal{P}_K = \{C_1^{(K)}, \ldots, C_{m_K}^{(K)}\}. \tag{23}$$

By varying K over a prescribed set of scales

$$K \in \{K_{\min}, K_{\min}+1, \ldots, K_{\max}\}, \tag{24}$$

while keeping the basal parameter K_0 fixed, one obtains a multiscale sequence

$$\mathcal{P}_{K_{\min}}, \mathcal{P}_{K_{\min}+1}, \ldots, \mathcal{P}_{K_{\max}}. \tag{25}$$

In this notation, K_0 denotes the basal neighbourhood scale from which the multiscale process is initiated, whereas K denotes the current layer scale. Thus, the layer sequence should be interpreted as a controlled sweep over K with K_0 kept fixed. This distinction is important in sparse-connectivity benchmarks, where an excessively large basal scale may prematurely merge regions connected by low-density bridges, while a smaller K_0 preserves the local structure long enough for bridge-like components to appear as separate clusters. Increasing K tends to increase graph connectivity. Therefore, small clusters from lower-K layers are expected to be absorbed into larger clusters at higher K values, although the exact behaviour depends on the dataset and on the visibility or leadership rules.

The overlap between clusters in consecutive layers is

$$W_{ab}^{(K,K')} = |C_a^{(K)} \cap C_b^{(K')}|. \tag{26}$$

Useful normalisations include

$$F_{ab}^{(K,K')} = \frac{|C_a^{(K)} \cap C_b^{(K')}|}{|C_a^{(K)}|}, \tag{27}$$

Table 1 Preliminary internal-overlap diagnostic for the variable-density blobs dataset. The equality between summed cluster sizes and number of unique points indicates that each layer is a valid partition.

K	number of clusters	summed size	unique points	duplicates
2	577	2000	2000	0
3	265	2000	2000	0
4	89	2000	2000	0
5	15	2000	2000	0
6	6	2000	2000	0
8	5	2000	2000	0

and the Jaccard index

$$J_{ab}^{(K,K')} = \frac{|C_a^{(K)} \cap C_b^{(K')}|}{|C_a^{(K)} \cup C_b^{(K')}|}. \tag{28}$$

8 Preliminary partition diagnostic

An important consistency check is whether each layer $\mathcal{P}_K$ is a true partition of the dataset. For each K, this can be verified by comparing the sum of cluster cardinalities with the cardinality of the union of all cluster members:

$$\sum_{a=1}^{m_K} |C_a^{(K)}| = \left| \bigcup_{a=1}^{m_K} C_a^{(K)} \right|. \tag{29}$$

When the equality in Eq. (29) holds and the union contains all N data points, the layer is not merely a cover of the dataset, but a proper partition:

$$\bigcup_{a=1}^{m_K} C_a^{(K)} = X, \qquad C_a^{(K)} \cap C_b^{(K)} = \varnothing \quad \text{for } a \neq b. \tag{30}$$

This property is important because the multiscale object produced by the algorithm is then a sequence of valid partitions, rather than a sequence of overlapping covers. Inter-layer overlap is still expected and informative, but intra-layer overlap would indicate an invalid layer representation.

If the left-hand side of Eq. (29) is larger than the right-hand side, at least one point belongs to more than one cluster in the same layer.

For the preliminary variable-density blobs example, the diagnostic gave the results in Table 1. Since all duplicate counts are zero, no internal overlap was detected within any analysed layer.

This diagnostic is different from inter-layer overlap. Inter-layer overlap is expected and informative; it describes how clusters from one scale are preserved, split, or absorbed at another scale.

9 Scale, transmissivity, and confidence

The older relational notes contained an interpretation in terms of transmissivity and confidence. In the present graph setting, this can be reinterpreted as a scale-selection heuristic. Small K values preserve local selectivity but may fragment the graph. Large K values increase propagation but can dilute local evidence and eventually merge distinct structures.

Let m denote the average number of forward links and n the average reverse-neighbour count. The old balance condition can be written abstractly as

$$\tau \approx \frac{mn}{MN}, \qquad \gamma \approx \frac{1}{mn}, \tag{31}$$

where τ measures propagation capacity and γ measures confidence or selectivity. The balance $\tau = \gamma$ yields

$$(mn)^2 = MN. \tag{32}$$

In a standard directed KNN graph, $m \approx K$, while the mean reverse degree is also approximately K. Therefore, the average alone is not sufficient; the distribution of reverse degrees is more informative. Quantities such as

$$\mathrm{Var}(\mathrm{rdeg}_K), \qquad \max_i \mathrm{rdeg}_K(x_i), \qquad H(\mathrm{rdeg}_K) \tag{33}$$

may help identify scales where local leadership is informative but not yet globally saturated.

10 Experimental protocol

The method should be evaluated on datasets that test distinct clustering difficulties:

- Gaussian blobs with equal density;
- blobs with variable density;
- two moons;
- concentric circles;
- anisotropic clusters;
- uniform or Poisson-like point clouds;
- clusters connected by sparse bridges;
- orbital or astronomical time-series embeddings.

The bridge benchmark plays a complementary role to moons and circles. While moons and concentric circles test nonconvex separation, the bridge dataset tests robustness under sparse connecting structures. It evaluates whether the method collapses two dense regions into a single component merely because they are connected by a low-density chain of points, or whether the bridge is preserved as a meaningful intermediate structure. This benchmark is therefore useful for studying premature saturation, scale sensitivity, and the effect of the basal parameter K_0.

When labels are available, external validation should include the adjusted Rand index and normalized mutual information [18, 19]. Internal criteria such as Davies–Bouldin, Calinski–Harabasz, and silhouette coefficients may also be reported [20–22], but should be interpreted carefully for nonconvex clusters. Baselines should include k-means, DBSCAN, HDBSCAN, SNN clustering, and mutual-KNN graph methods when available.

For each dataset and each K, the following quantities should be reported:

- number of clusters m_K;
- cluster size distribution;
- saturation depth statistics;
- reverse-degree distribution;
- ARI and NMI when labels are known;
- internal validity indices;
- layer-transition overlaps $W_{ab}^{(K,K')}$ and normalized quantities $F_{ab}^{(K,K')}$ and $J_{ab}^{(K,K')}$.

11 Discussion

The method should be positioned carefully. It is not a first use of KNN graphs for clustering, nor a replacement for all density-based methods. Its distinctive aspect is the recursive use of forward KNN selection followed by reverse-neighbour retrieval, with popularity-based leadership and saturation. This places the method between shared-neighbour clustering, reverse-neighbour density reasoning, and graph-closure approaches.

The method is intentionally presented here as an algorithmic and topological construction, rather than as the minimisation of an explicit objective function. Its definition does not reduce to a weighted voting rule, nor to the particular choice of a distance-based weight such as an inverse-square law. The central object is the dynamics induced on the directed KNN graph: forward-neighbour selection, reverse-neighbour retrieval, popularity, visibility, recursive expansion, and saturation. Whether this process admits an equivalent variational, probabilistic, or energy-based interpretation remains an interesting question for future work, but the absence of such a formulation is not a limitation of the graph operator itself.

Several points remain to be specified rigorously before a final submission:

1. the exact closure rule used by the reference implementation;
2. how ties in popularity leadership are handled;
3. whether visibility constraints are mandatory or optional;
4. how saturated closures are merged into final clusters;
5. the computational complexity of the closure process;
6. robustness to noise and outliers;
7. behaviour on homogeneous data, where saturation may correctly produce a single large component.

12 Conclusion

We presented a draft formulation of a clustering method based on round-trip closure in directed KNN graphs. The method combines a relational forward-backward idea, reverse-neighbour popularity, local leadership, recursive expansion, saturation, and multiscale K-layer analysis. The 2013 relational model provides the mathematical origin of the reflection operator, while the 2018 KNN prototype provides the algorithmic core based on reverse-neighbour discovery and popularity leadership. The modern formulation casts these ideas as a graph-based clustering variant suitable for systematic validation against established nearest-neighbour and density-based clustering methods.

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- Materials availability: Not applicable.
- Code availability: The implementation will be made available in a public repository upon publication. To be completed.
- Author contribution: To be completed.

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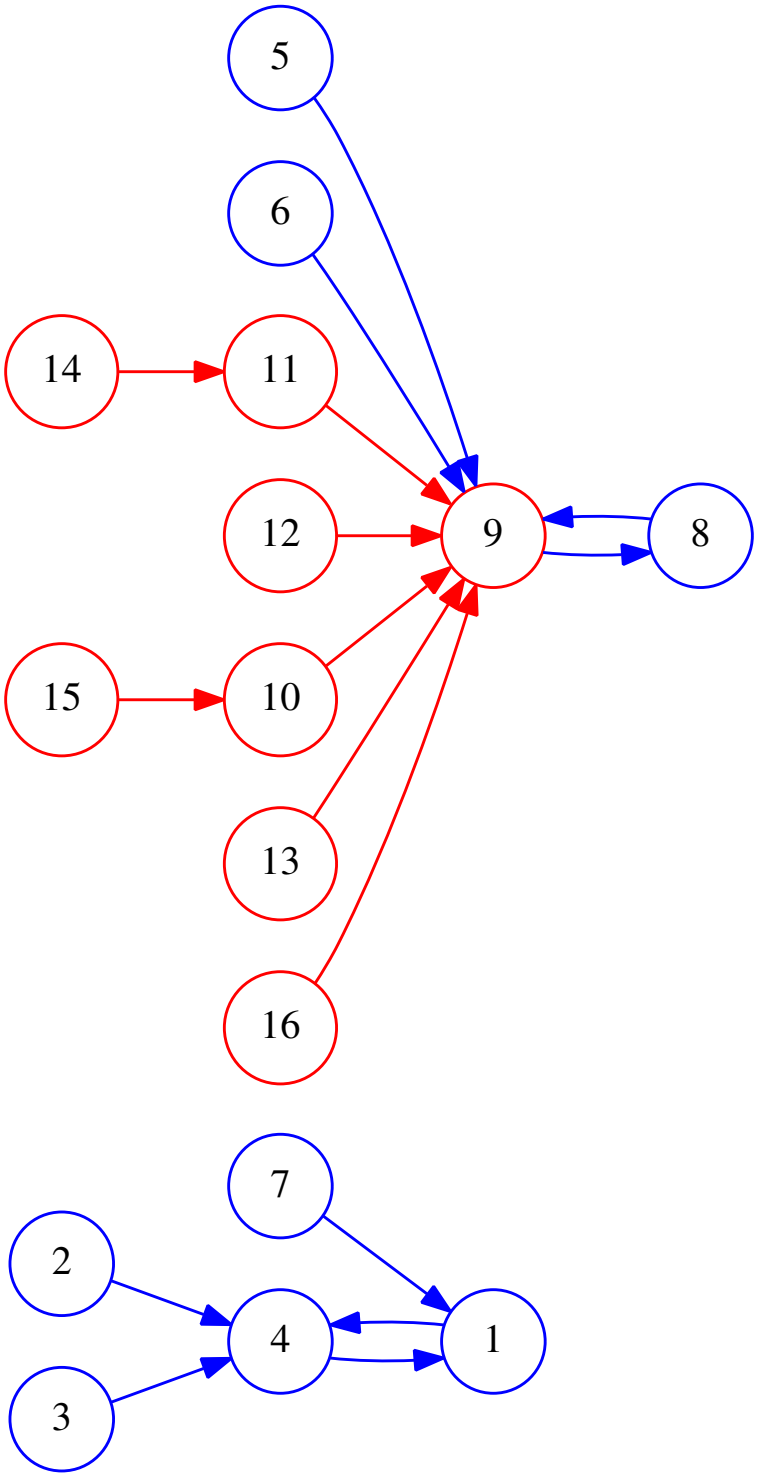


Fig. 3 Popularity-based leadership graph from the 2018 prototype. Each point points to a locally selected leader among its nearest neighbours according to reverse-neighbour popularity.

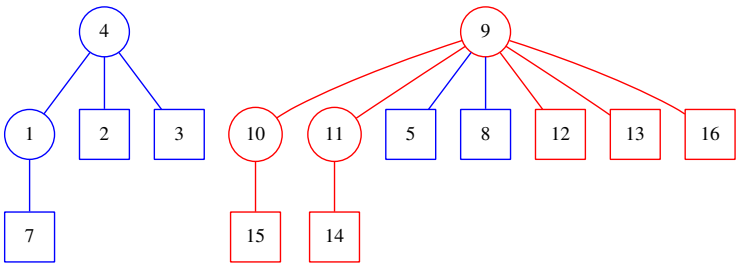


Fig. 4 Cluster-tree representation from the 2018 prototype. In the present formulation, this structure is reinterpreted as an ancestor of the K-layer transition graph, where clusters at smaller K are progressively absorbed by larger-scale structures.

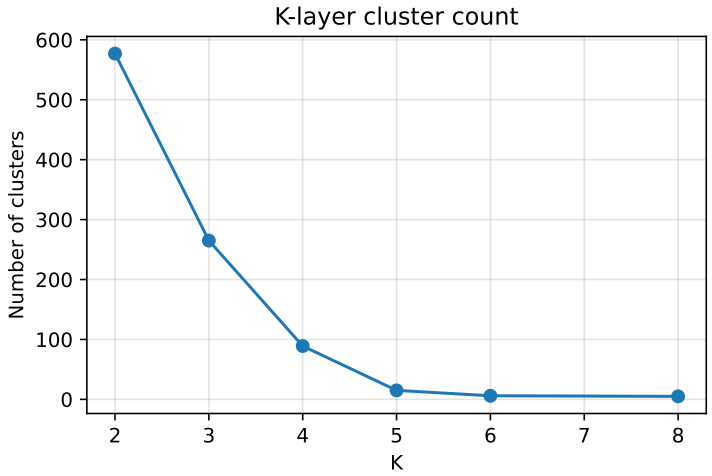


Fig. 5 Preliminary K-layer evolution for a variable-density blobs dataset with $N = 2000$ points. The number of clusters decreases as K increases, indicating progressive absorption of microclusters into larger structures.